Database Change Management Procedure

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Summary. Procedure for planning, reviewing, testing and applying database schema and data changes at Northstar Systems. Covers migration scripts, DBA review, expand and contract for Tier-1 services, destructive change rules, deployment windows, data fixes, rollback and emergency handling.

SYNTHETIC DOCUMENT - Northstar Systems is a fictional company. This document exists for development, testing and demonstration only and is not real legal, employment, security or compliance advice.

1. Purpose and Scope

This procedure governs every change to the structure or content of databases that support Northstar Systems Private Limited products and internal systems: schema changes (tables, columns, indexes, constraints, views, stored procedures), reference data changes and one-off data fixes. It applies to the databases behind the Northstar Nexus Platform, Nexus Connect and Nexus Insight, and to internal systems operated by Engineering, in all environments, with the strictest controls applied to production.

The aim is that no database change causes customer-visible downtime, data loss or corruption, that every change is traceable and reviewable, and that every change can be recovered from. The procedure complements the deployment runbook ENG-003, the pipeline reference ENG-007, the disaster recovery and backup runbook ENG-014 and the data protection controls in SEC-005. Where a database change forms part of an incident response, ENG-004 also applies.

2. Definitions

- **Migration script**: a versioned, ordered script, stored in the repository, that moves a database from one schema version to the next.
- **Schema change**: any change to the structure of a database.
- **Data fix**: a manual or scripted change to data rows in production, not part of normal application behaviour.
- **Expand and contract**: a pattern in which a change is made in two or more releases. First the schema is expanded in a backwards-compatible way, then the code is moved to use it, and only later is the old structure removed (contracted).
- **Destructive change**: a change that removes or irreversibly alters structure or data, such as dropping a table or column, narrowing a data type, or deleting rows.

- **Data Engineering DBA**: the database administrator in the Data Engineering team who reviews migrations.
- **Tier-1, Tier-2, Tier-3 services**: customer-facing and revenue critical, internal critical and non-critical services respectively.
- **Restricted data**: customer production data, personal data of employees, credentials and similar classified data that requires encryption and tight handling.
- **Deployment window**: Tuesday to Thursday, 10:00 to 16:00 IST for production.

3. Roles and Responsibilities

Role	Responsibilities
Change author (developer)	Writes the migration script, tests it, prepares the rollback or forward-fix plan, raises the NorthDesk change ticket
Data Engineering DBA	Reviews every schema migration, checks performance and locking impact, approves for production
Second reviewer	A second engineer who reviews data fixes and Tier-1 migrations
Service owner (engineering manager)	Accepts the risk of the change and confirms timing
Release manager	Schedules the migration with the release train and verifies pipeline gates
SRE	Confirms backup status before the change, monitors during and after
Head of Data Engineering	Owns this procedure and approves exceptions
Security Champion	Reviews changes affecting personal data columns, encryption or access rights
Privacy Officer	Consulted when a change alters how personal data is stored or retained

4. Principles

1 All schema changes must be made only through migration scripts in version control. Manual schema changes in production using a console are prohibited.
2 Every migration is reviewed by the Data Engineering DBA before it can be merged.
3 Changes to Tier-1 databases must be backwards-compatible, using the expand and contract pattern.
4 A destructive change must never be made in the same release as the code that stops using the structure.
5 Production migrations may be run only inside the deployment window.
6 Data fixes in production need two-person review.
7 Every production change needs a NorthDesk change ticket and a rollback or forward-fix plan, as required for all production changes.

5. Migration Scripts

5.1 Standards

- Each migration is a numbered, immutable script such as `V2026_03_001__add_pipeline_owner_column`. Once merged, a migration is never edited; corrections are made in a new migration.

- Migrations are idempotent where possible and run inside a transaction if the database supports it.
- Scripts must not contain credentials or personal data. Reference data scripts use identifiers, not real customer values.
- Large changes must be batched to avoid long locks. Typical batch size should be chosen by the DBA, and a script that updates more than 1 million rows must include a throttle.
- Index creation on large tables must use online or concurrent creation methods.
- Every migration includes a header comment with the ticket number, author, summary, expected duration and whether it is expand, contract or neutral.

5.2 Review checklist for the Data Engineering DBA

1 Does the script conform to naming and numbering?
2 Is it backwards-compatible with the currently deployed code (Tier-1 mandatory)?
3 What locks does it take, and for how long on production data volume?
4 Are indexes and constraints consistent with query patterns?
5 Does it touch personal data columns? If yes, has the Security Champion reviewed it?
6 Is there a tested rollback or a documented forward-fix?
7 Has it been tested on a staging copy of realistic size?
8 Is it destructive? If yes, is it separated from the code release as required?

The DBA must give the first response to a review request within 1 business day, in line with the first review response target in ENG-001.

6. Expand and Contract

For Tier-1 services every schema change must be backwards-compatible so that the old and new versions of the code can run against the same database at the same time during the rollout (canary 10%, 50%, then 100%) and after a rollback.

Phase	What is done	Release
Expand	Add new columns, tables or indexes as nullable or with defaults. Old code ignores them.	Release 1
Migrate	Deploy code that writes to both old and new structures, backfill existing data in batches	Release 2
Switch	Deploy code that reads only the new structure	Release 3
Contract	Remove the old structure after verifying it is unused	Release 4 or later

The contract step must be a separate release from the code that stops using the old structure. A waiting period of at least one full release train between the switch and the contract step is recommended, so that a rollback of the code remains safe.

Example: The team wants to rename the column `owner` to `owner_id` in a Tier-1 table. Instead of a rename, they add `owner_id` in Release 1 (expand), deploy code that writes both and backfills in Release 2, deploy code that reads `owner_id` in Release 3, and drop `owner` in Release 4, which contains no code that uses `owner`. If Release 3 is rolled back, the old code still finds `owner` populated.

Tier-2 and Tier-3 services should use the same pattern, but the Data Engineering DBA may accept a single-release change where a short maintenance window is agreed in advance with affected

internal users.

7. Destructive Changes

A destructive change must never be deployed in the same release as the code that stops using the structure. The reason is rollback safety: if the code is rolled back after the structure is dropped, the application will fail.

Additional rules:

1 A full backup must have completed successfully before the change. For Tier-1 databases, daily full backups and continuous 15-minute log shipping are in place as described in ENG-014, and the DBA must confirm the latest backup time before the change starts.
2 The data owner and the service owner must approve in the NorthDesk ticket.
3 Dropped columns and tables should be first renamed or hidden and kept for at least one release cycle before final deletion, where storage permits.
4 Deleting customer data must follow retention rules and legal holds. A legal hold overrides retention and blocks deletion. The Privacy Officer must be consulted if the deleted data includes personal data.
5 Deletes of more than 10,000 rows in production must be run in batches with progress monitoring, and require DBA presence.

8. Production Execution

8.1 Window and schedule

Production migrations are only allowed in the deployment window, Tuesday to Thursday, 10:00 to 16:00 IST, and not during freeze periods (the last 3 business days of each quarter and company-declared freezes). They are scheduled with the weekly release train, which departs on Tuesday. A migration that is part of a Normal change must be on the change request by Monday 12:00 IST for the Monday 15:00 IST CAB meeting.

8.2 Steps

1 Confirm the NorthDesk change ticket has the migration reference, DBA approval, rollback or forward-fix plan, and expected duration.
2 Confirm with SRE that the latest backup and log shipping are healthy and replication lag is normal.
3 The release manager announces the start in the release channel.
4 The pipeline applies the migration through NorthShip using a migration service account with the minimum privileges needed. Engineers do not run migrations by hand from their laptops.
5 The DBA or SRE monitors locks, replication lag, error rates and latency in NorthWatch during the migration.
6 After completion, validation queries are run and results are recorded in the ticket.
7 The release proceeds with the canary and staged rollout only after the migration check passes.
8 The ticket is closed after 24 hours of stable operation for Tier-1.

8.3 Abort criteria

The change author or DBA must stop the migration if any of the following occur: replication lag grows beyond the agreed threshold, locks are held longer than expected, error rate rises above 2%, p95 latency exceeds 1.5x baseline for 5 consecutive minutes, or the migration runs more than

50% over its expected duration. The decision to abort is made by the DBA and recorded in the ticket.

9. Data Fixes

A data fix is any production data change made outside normal application operation, for example correcting records after a defect.

1 The requester raises a NorthDesk ticket explaining the issue, affected rows, the query and the expected outcome.
2 The author writes the script and a verification query, and runs it first against a staging copy where feasible.
3 The data fix needs two-person review: a second engineer, who must not be the author, reviews the script and the target data set, and the DBA approves for production.
4 Before running, the author takes a backup or snapshot of the affected rows into a restricted location and records its location.
5 The fix runs inside the deployment window except for an Emergency change.
6 The script runs through the approved production access route, the NorthGate bastion, with session recording. Privileged access is just-in-time for at most 8 hours and needs Director approval and CISO delegate approval.
7 After the run the author records the number of rows affected, runs the verification query and attaches both to the ticket.

Example: A defect created 312 duplicate subscription records. The author prepares a delete script restricted by ticket reference, a second engineer reviews it and confirms the 312 rows, the DBA approves, the script is run on Wednesday at 11:00 IST, and the row count of 312 is recorded as expected.

Data fixes touching personal data must be reported to the Privacy Officer, and the fix must be minimal and logged.

10. Rollback and Recovery

Because some changes cannot be reversed, each change ticket must state one of the following.

- **Rollback**: a tested script that returns the database to the previous structure without data loss. Allowed for expand changes.
- **Forward-fix**: a plan to fix issues with a new migration, used when rollback would lose data.
- **Restore**: recovery from backup, used only as a last resort because it can lose data written since the backup. For Tier-1 databases the recovery objectives are RPO 15 minutes and RTO 1 hour, as in ENG-014.

Restore tests of Tier-1 backups are run quarterly, and full disaster recovery drills happen twice a year, in April and October. Backups are encrypted and copied to the secondary region Hyderabad. The change author does not run restores; they request them through SRE and the incident process in ENG-004.

11. Security and Compliance

- Production database credentials are stored in NorthVault and rotated: service account credentials every 90 days.
- Data at rest is encrypted with AES-256 and data in transit with TLS 1.2 minimum.

- Direct developer access to production databases is not allowed other than through the just-in-time route described above. All privileged actions are logged and retained for 24 months in tamper-evident storage.
- Columns containing personal data must be identified in the schema documentation, and new personal-data columns need a Data Protection Impact Assessment when the processing is high risk (illustrative; to be verified by Legal before external use).
- Test and staging databases must not contain real customer production data, per ENG-008.

12. Exceptions and Escalation

- Emergency schema or data changes during an incident follow the Emergency change route: approval by the Engineering Director on call, a retrospective within 1 business day, and a DBA review as soon as available, within 1 business day. Emergency changes must still follow the destructive change rule where possible.
- If the Data Engineering DBA is unavailable, the Head of Data Engineering names a delegate. A Tier-1 migration must not proceed without a DBA or delegate.
- A Tier-1 change that cannot be made backwards-compatible (for example a cryptographic key migration) requires a written exception from the Head of Data Engineering and the VP Engineering, a maintenance announcement on Northstar Status at least 7 days ahead, and a window within 01:00 to 05:00 IST agreed in advance.
- Disagreements between author and DBA are escalated to the Head of Data Engineering within 1 business day.

13. Frequently Asked Questions

Can I rename a column in a Tier-1 table in one release? No. Use expand and contract so old and new code both work during rollout and rollback.

Can I drop the old column in the same release that stops using it? No. A destructive change must not ship with the code that stops using the structure.

Can I run a migration on Friday? No. Production migrations run only in the Tuesday to Thursday deployment window, apart from Emergency changes.

Who can run a data fix in production? The change author, after two-person review, DBA approval and just-in-time privileged access.

14. Related Documents

- ENG-003 Software Deployment Runbook: deployment and rollback rules.
- ENG-014 Disaster Recovery and Backup Runbook: backups, RPO and RTO and restore tests.
- SEC-005 Data Classification and Handling Policy: handling of Restricted data.
- ENG-007 CI/CD Pipeline Reference: pipeline execution of migrations.
- ENG-004 Production Incident Management Procedure: incidents arising from database changes.

15. Revision History

Version	Date	Change
4.0	2025-03-04	Made expand and contract mandatory for Tier-1 services
4.1	2025-06-24	Added data fix two-person review and abort criteria

Version	Date	Change
4.2	2025-10-26	Clarified destructive change separation and emergency handling